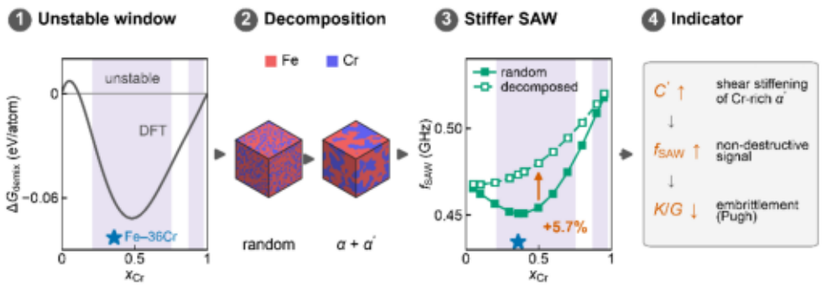
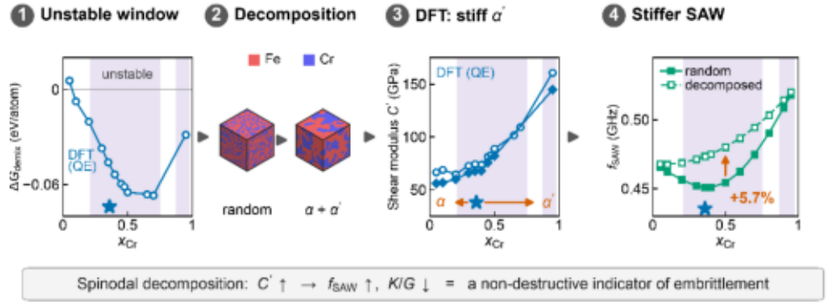
# Graphical abstract: all versions (newest last)

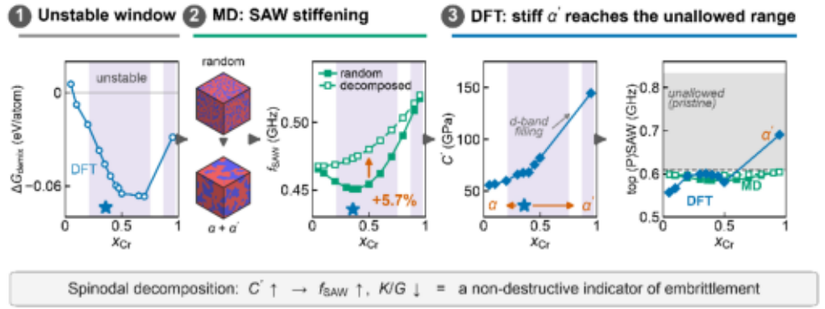
## v1



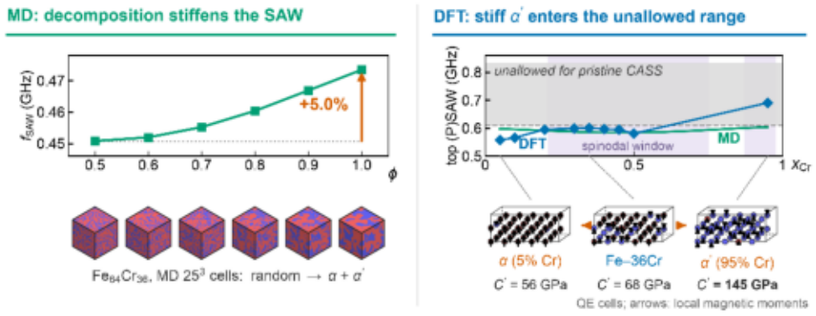

## v2



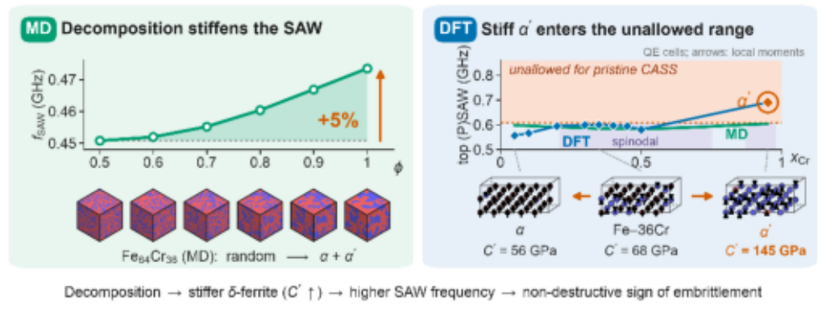

## v3



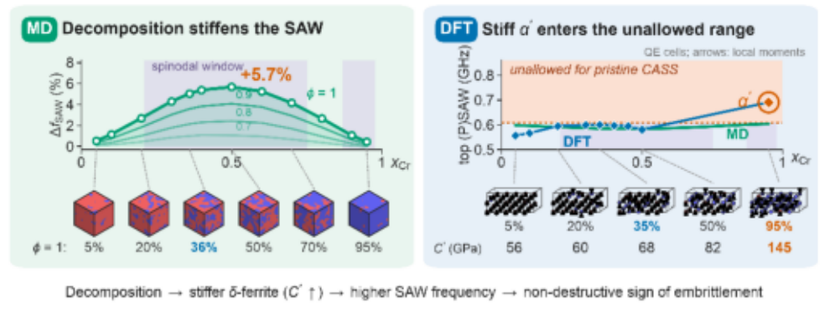

## v4



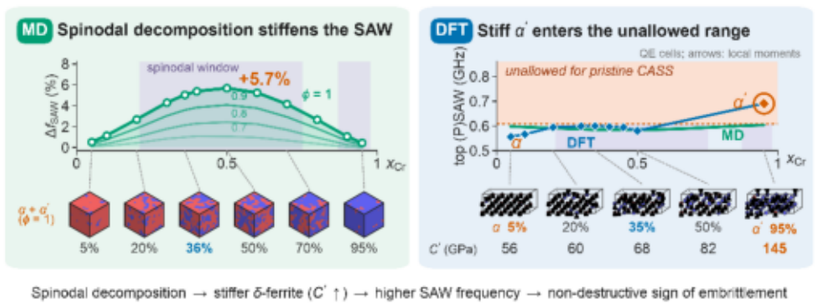

## v5



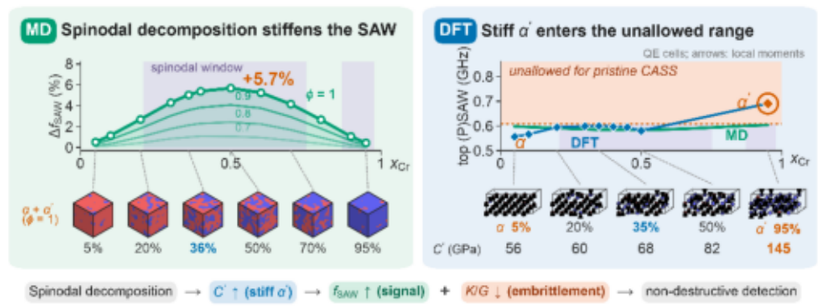

## v6



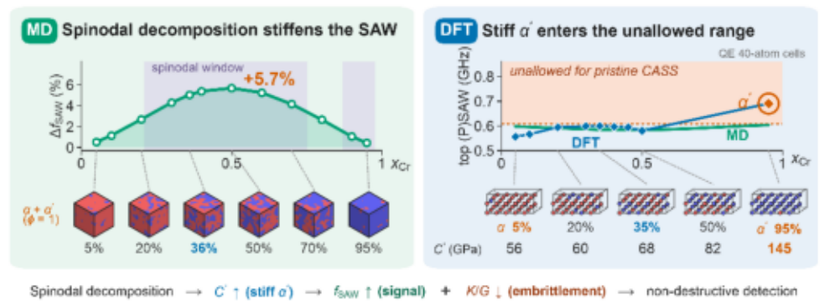

## v7



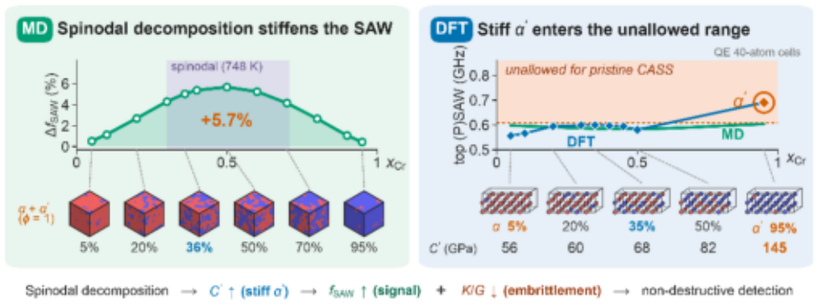

## v8



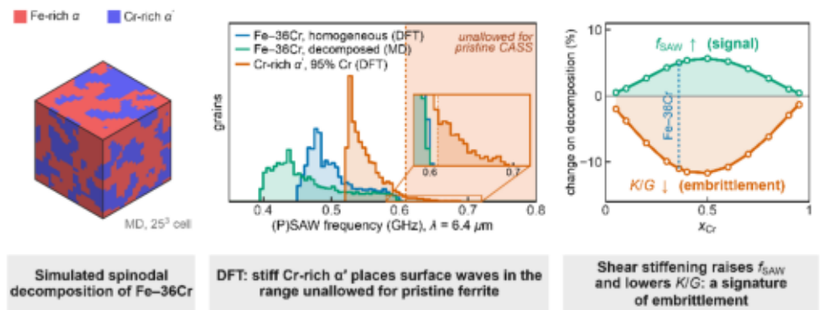

## v9



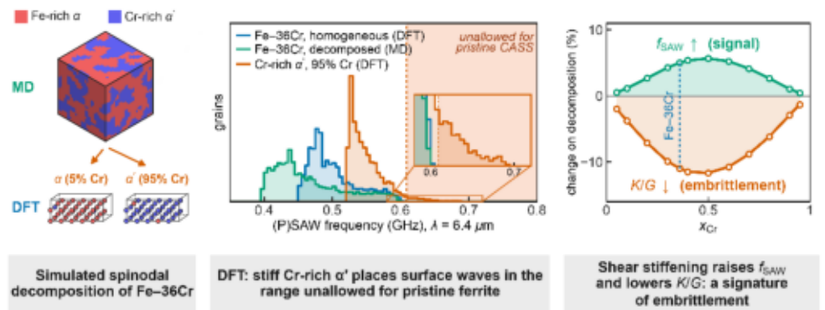

## v10



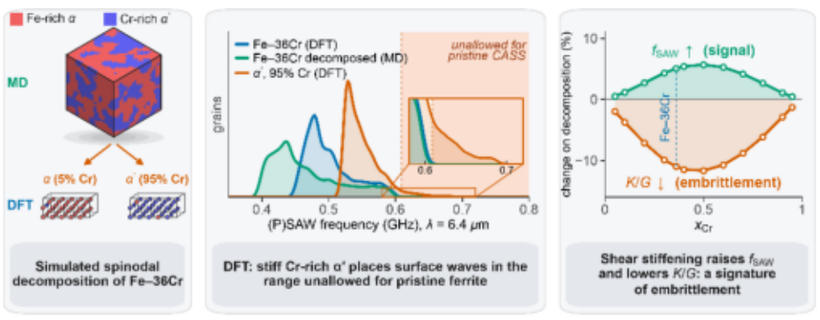

## v11



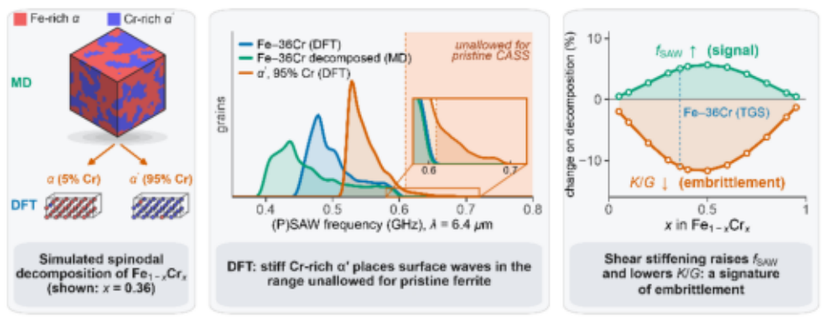

## v12



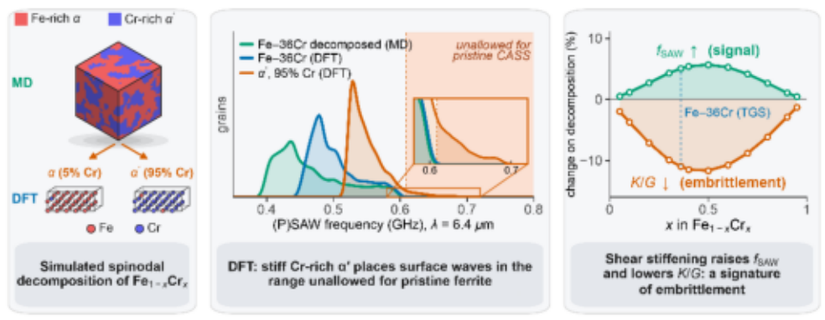

## v13



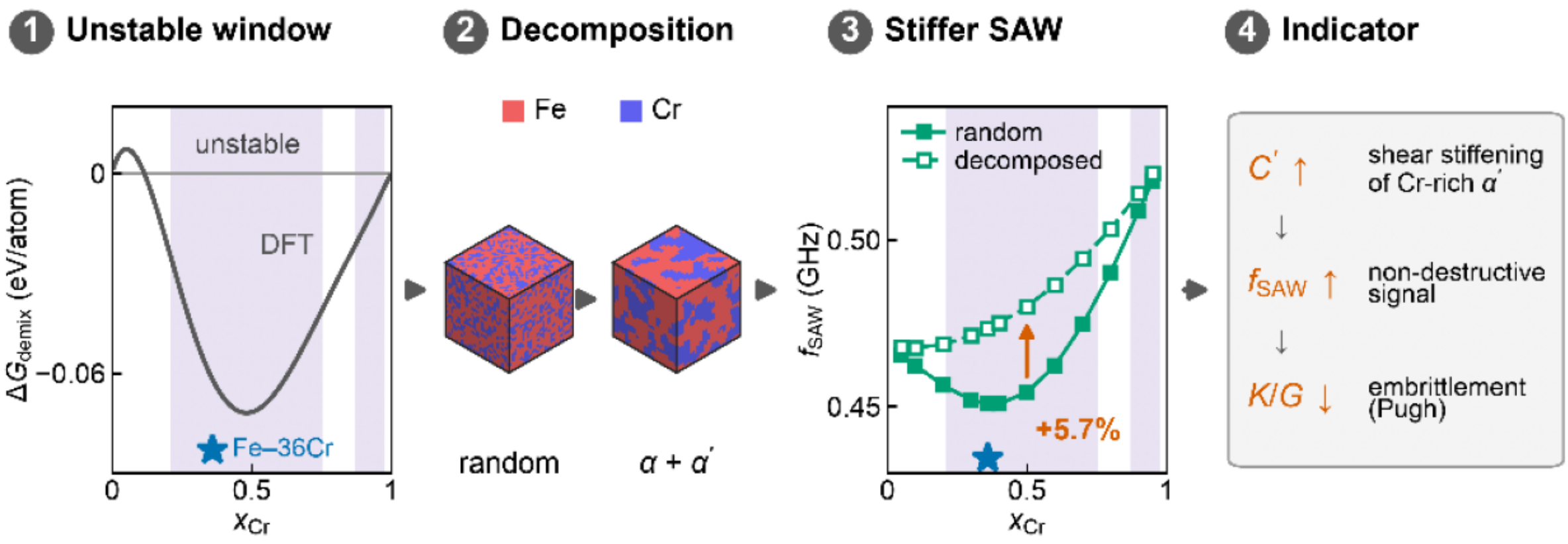

## v14




## v15

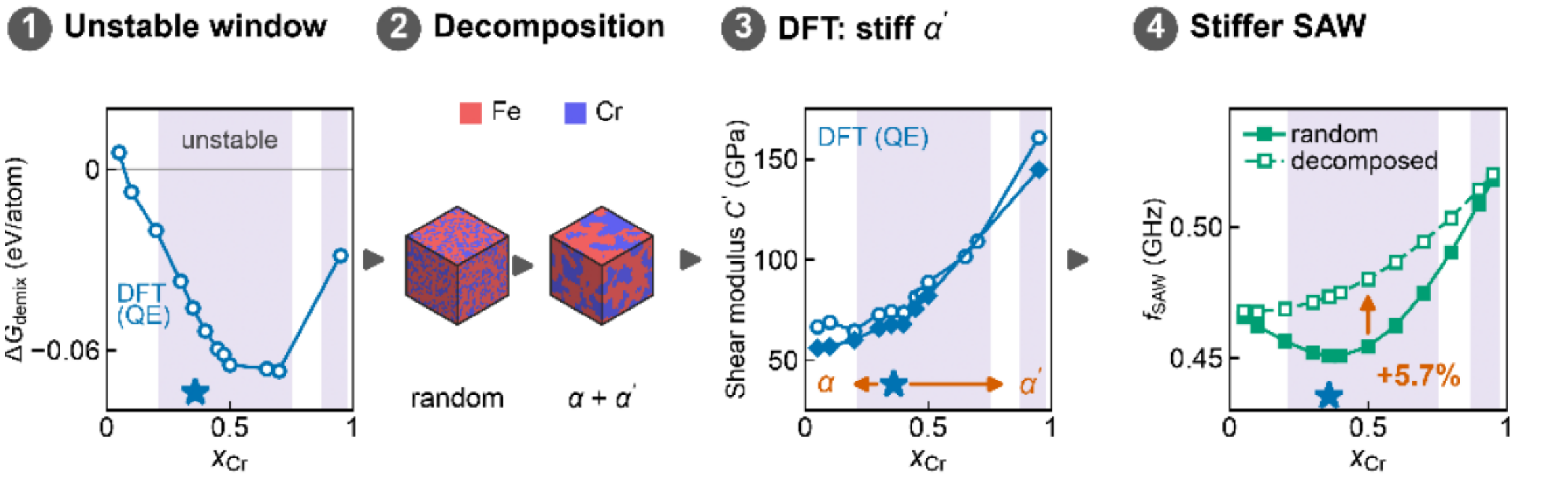


**v1**




First draft: four numbered steps (unstable window, MD cubes, SAW stiffening, indicator box)

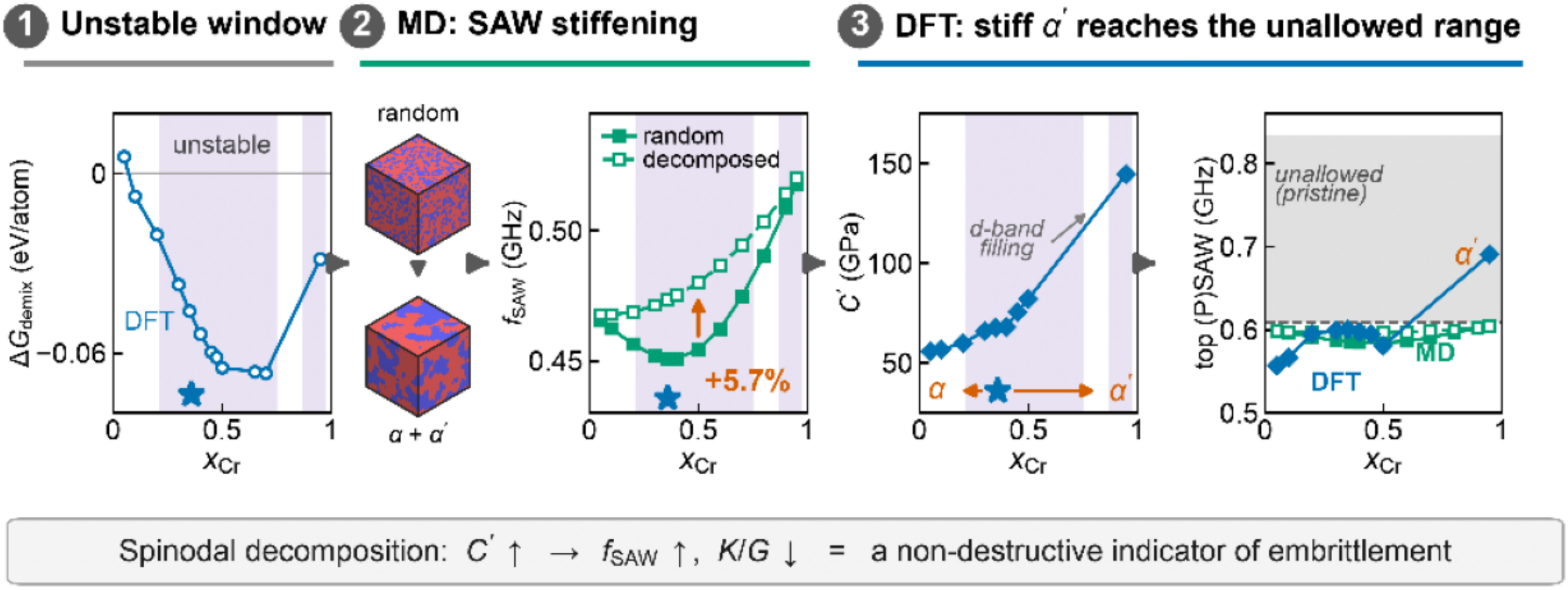
**v2**




Spinodal decomposition: $C' \uparrow \rightarrow f_{SAW} \uparrow$, $K/G \downarrow$ = a non-destructive indicator of embrittlement

DFT added: QE demixing energy (1) and QE C' (3); conclusion as banner

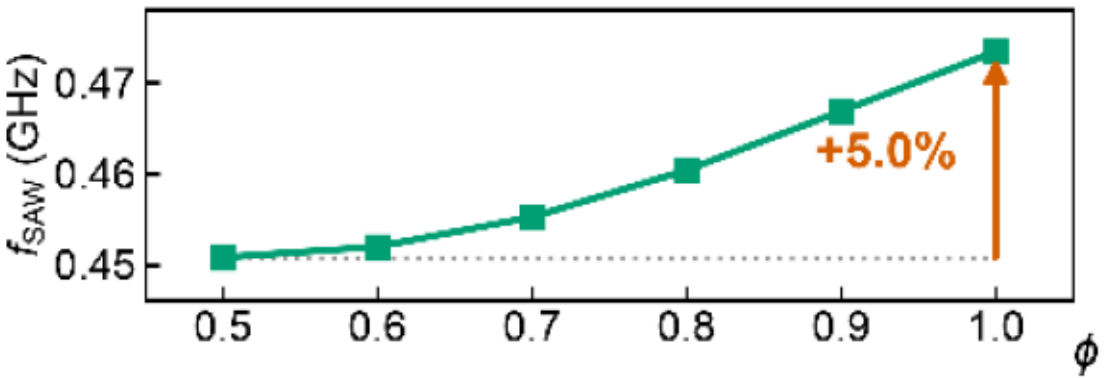
**v3**

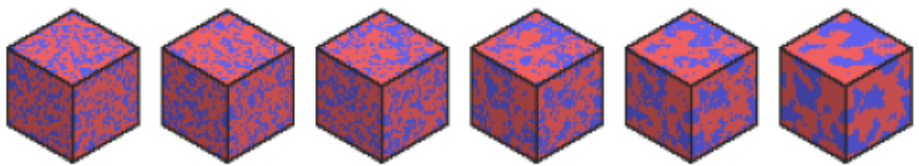



MD and DFT side by side; DFT top (P)SAW entering the unallowed range (Fig. 2b)

**v4**

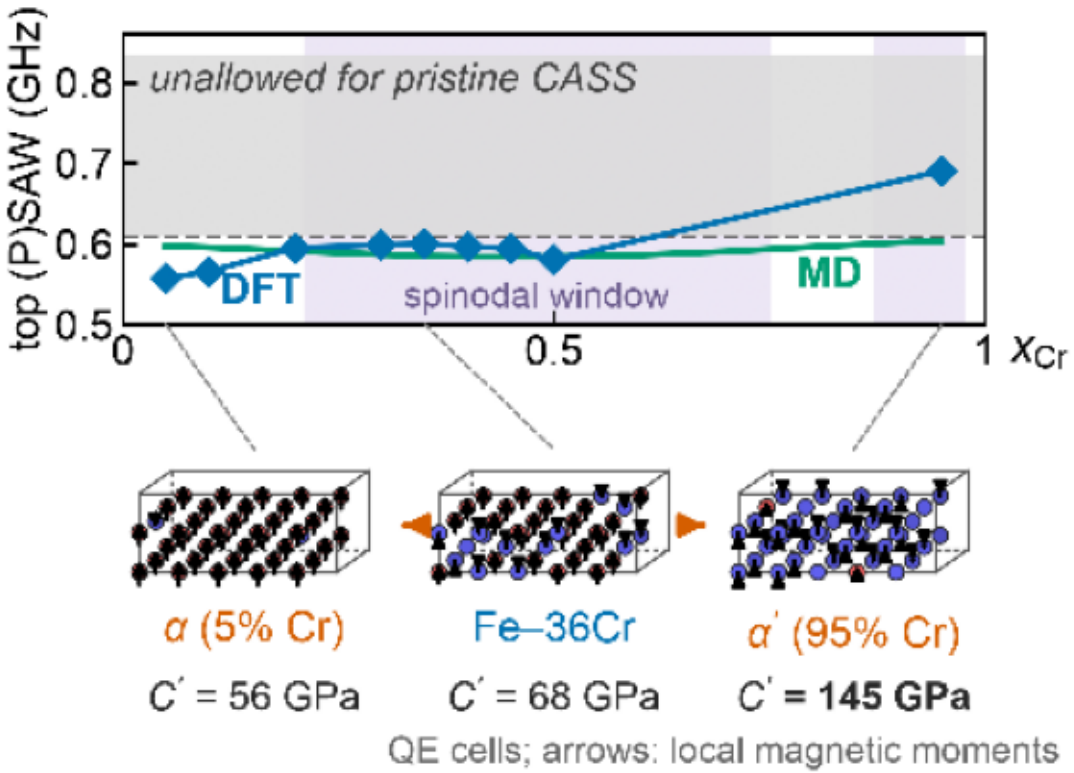
## MD: decomposition stiffens the SAW



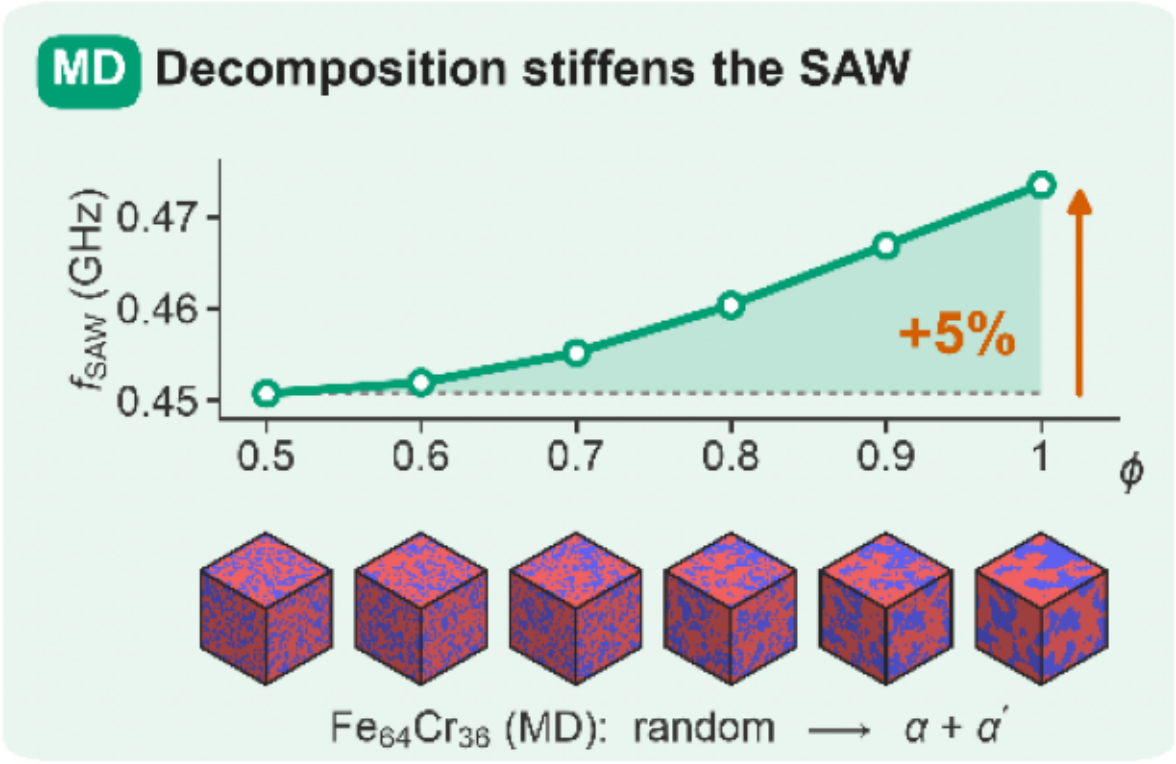
$Fe_{64}Cr_{36}$, MD $25^3$ cells: random → $\alpha + \alpha'$

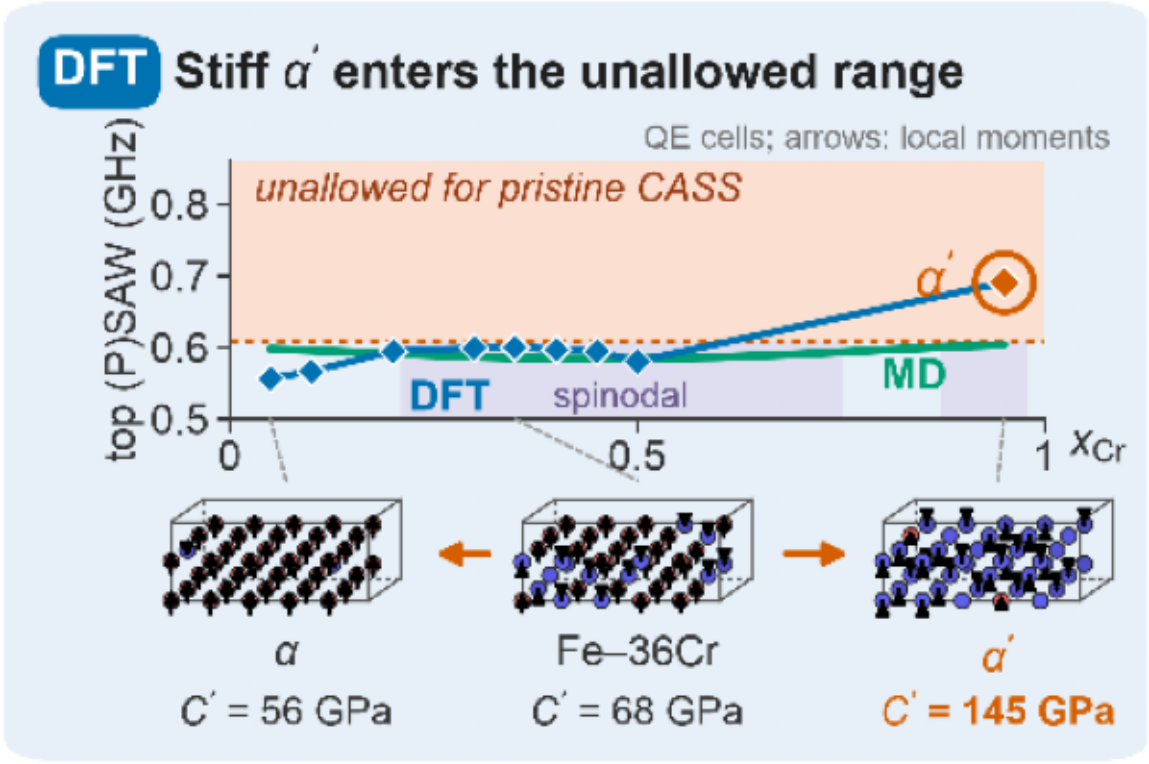
## DFT: stiff $\alpha'$ enters the unallowed range




Two halves: key result plotted above its structures (MD cubes along phi; QE cells alpha, Fe-36Cr, alpha')

# v5

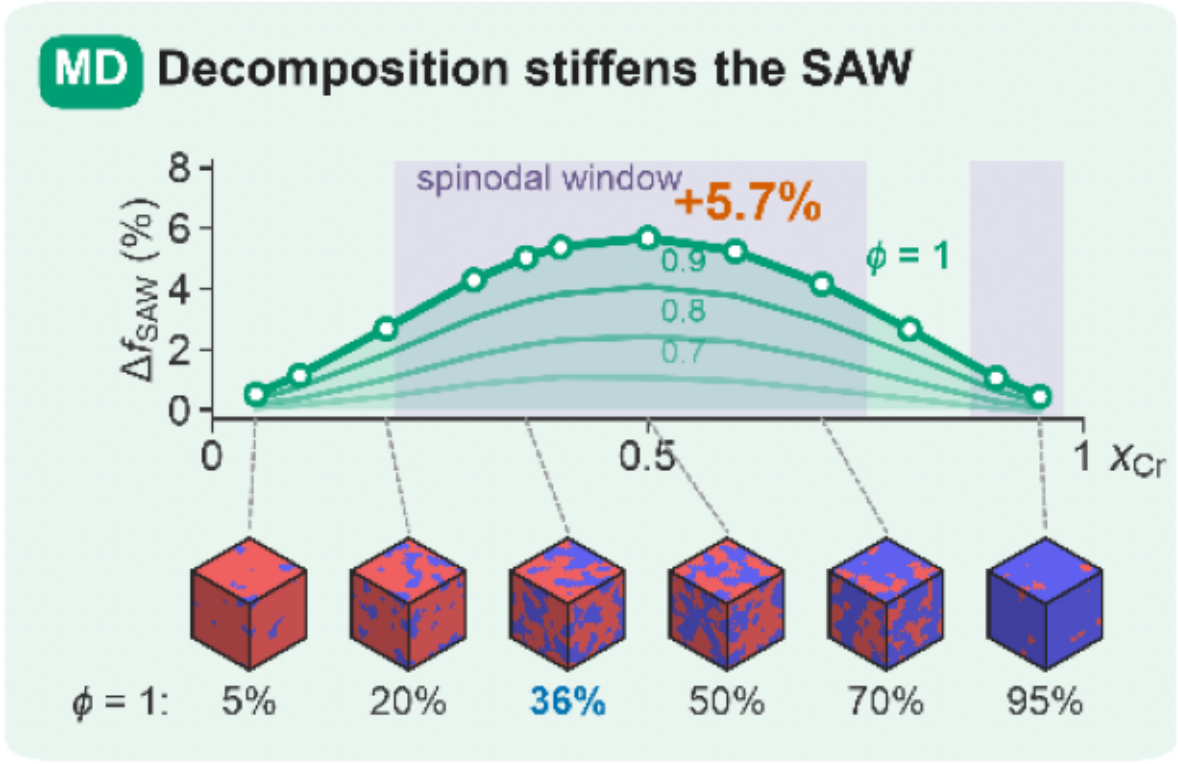


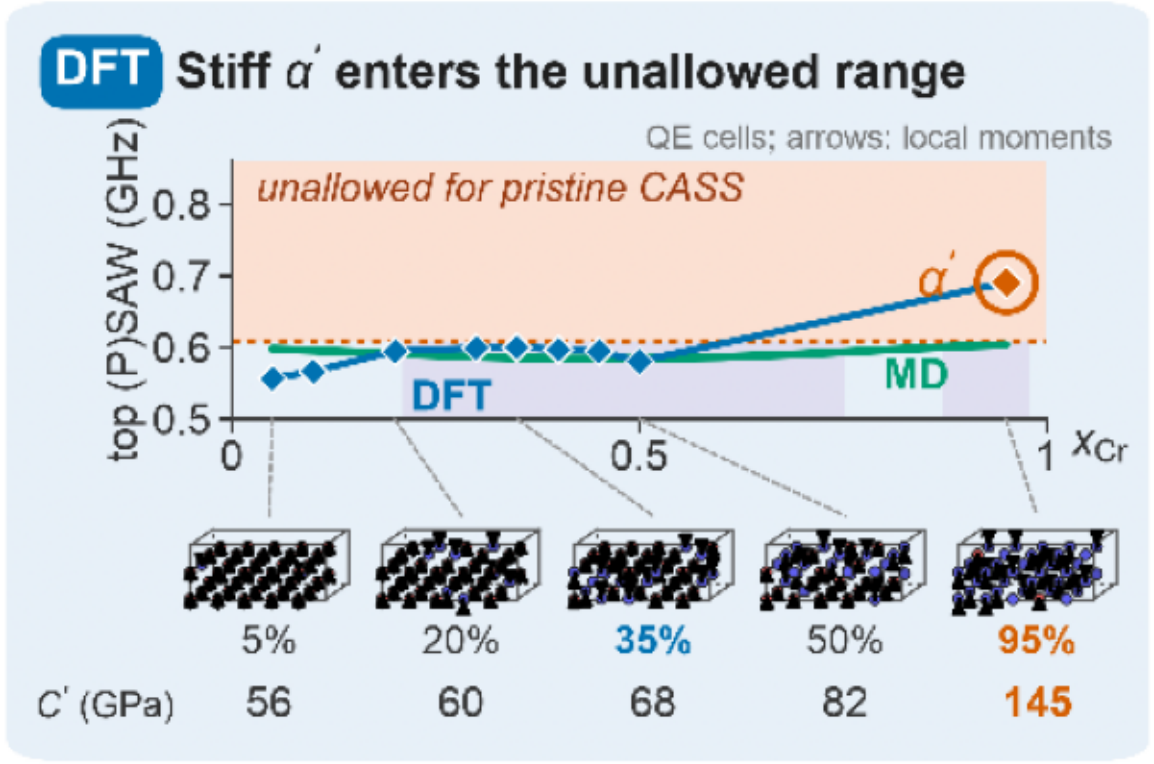




Decomposition → stiffer δ-ferrite ($C'$ ↑) → higher SAW frequency → non-destructive sign of embrittlement

v4 restyled: tinted cards, MD/DFT tags, open axes, highlighted alpha', banner

# v6

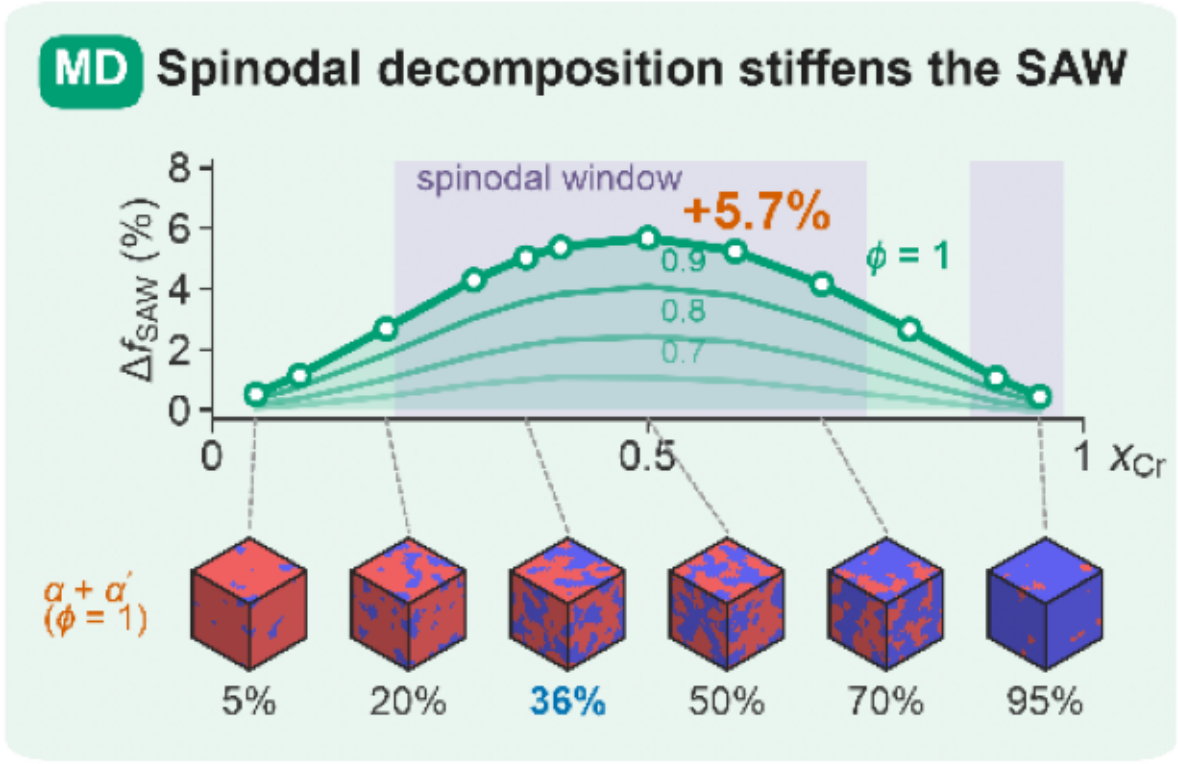


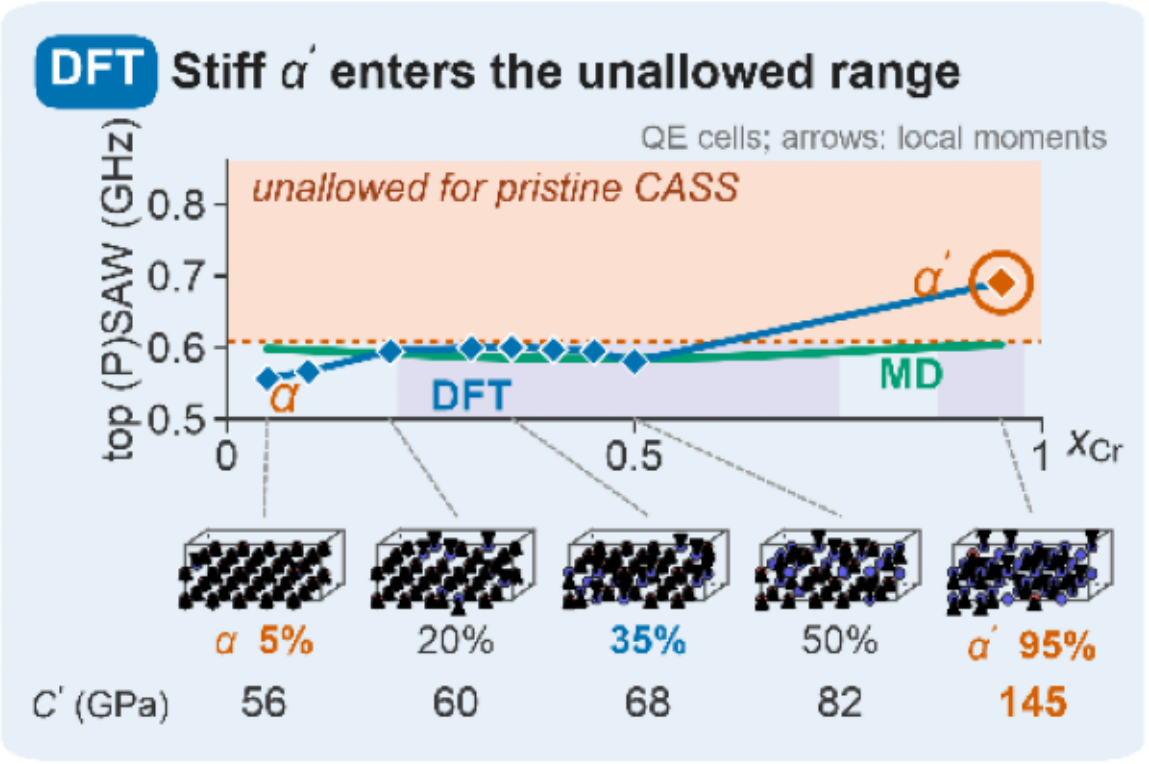




Decomposition → stiffer δ-ferrite ($C'$ ↑) → higher SAW frequency → non-destructive sign of embrittlement

Both cards on one x_Cr axis: all twelve MD compositions (phi = 0.7-1), six MD cells, five QE cells with C'

**v7**

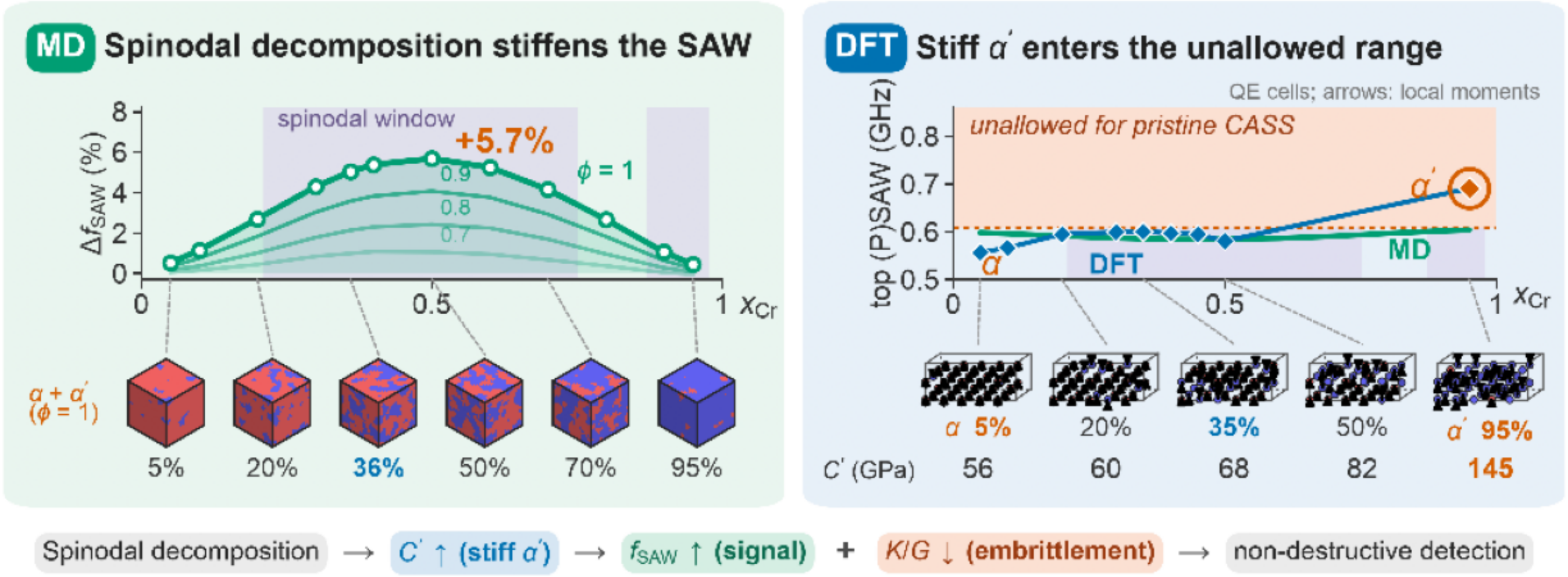



Spinodal decomposition → stiffer δ-ferrite ($C'$ ↑) → higher SAW frequency → non-destructive sign of embrittlement

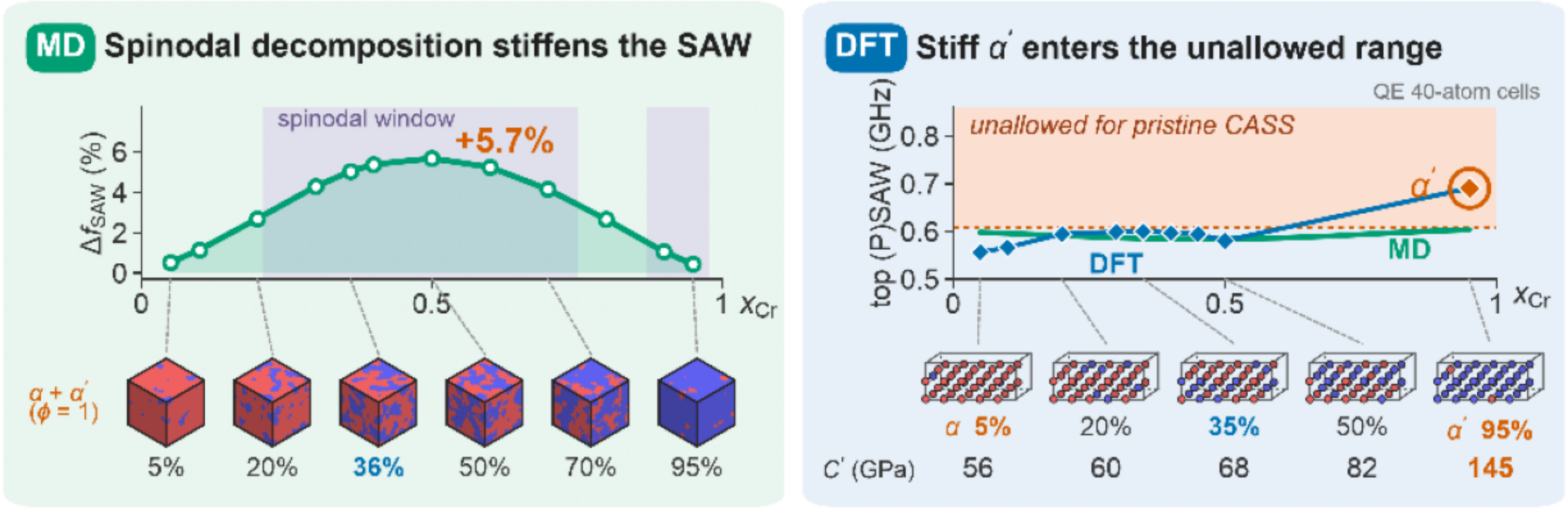
"Spinodal decomposition" in title and banner; alpha / alpha' labels back

**v8**




Mechanism line as coloured chips including K/G down (embrittlement)

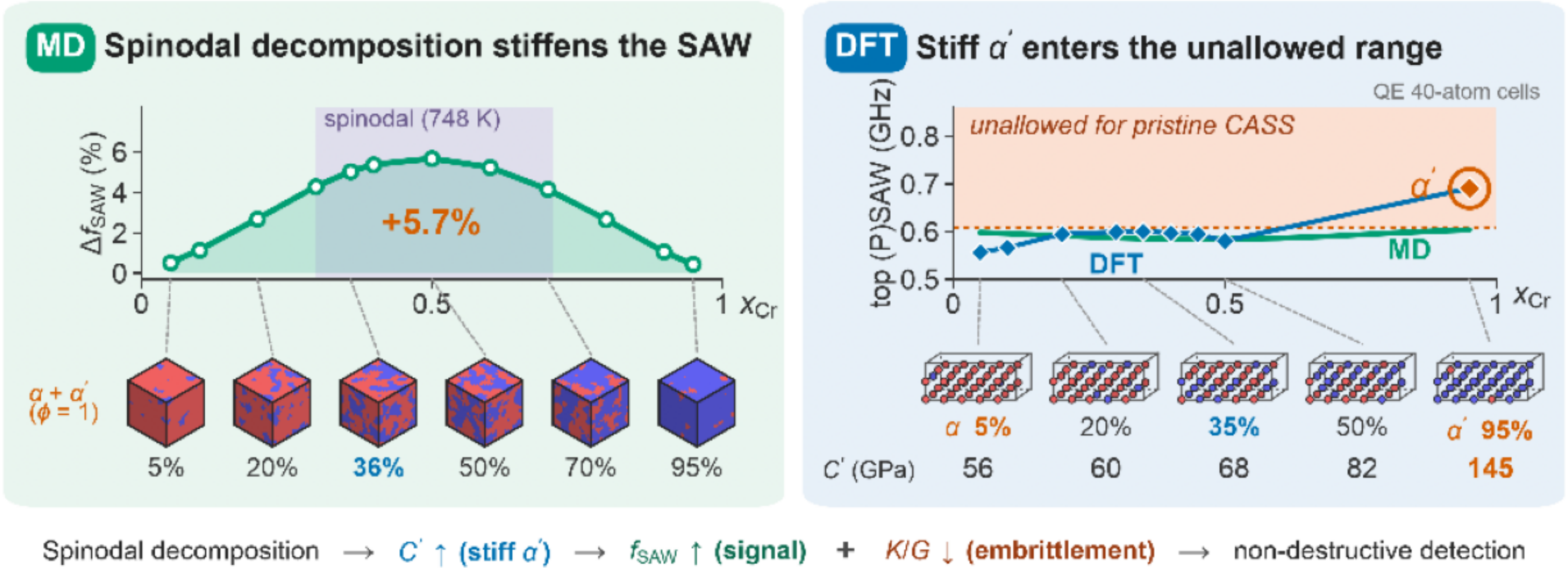
# v9




Spinodal decomposition → $C'$ ↑ (stiff $\alpha'$) → $f_{SAW}$ ↑ (signal) + $K/G$ ↓ (embrittlement) → non-destructive detection

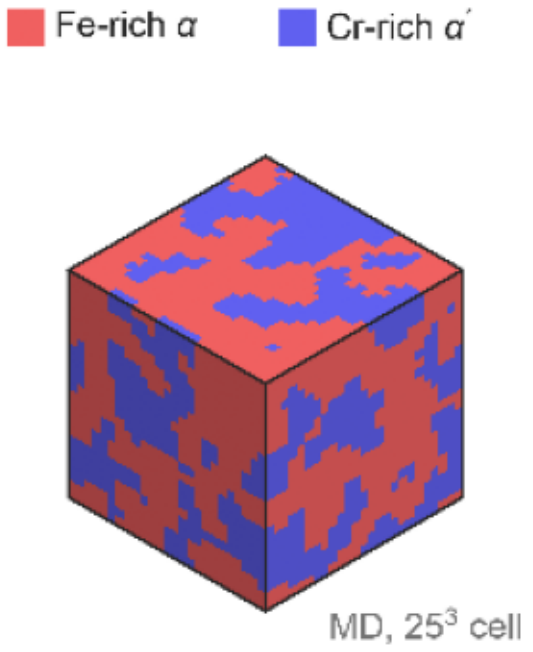
Decluttered v8: phi = 1 curve only, QE cells without moment arrows, no window shading or extra alpha label in the DFT plot, mechanism line as plain coloured text

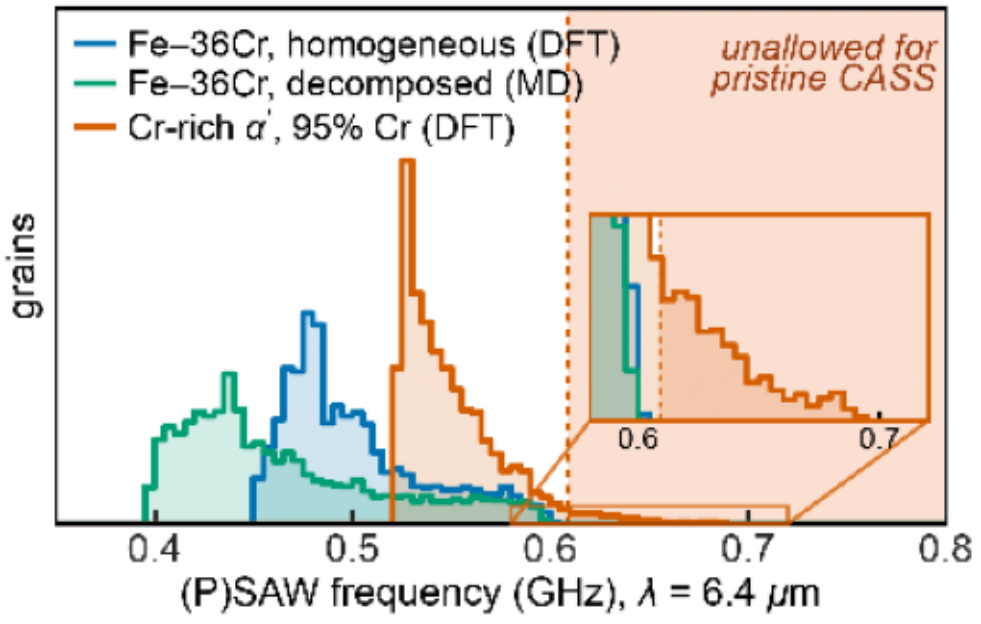
**v10**

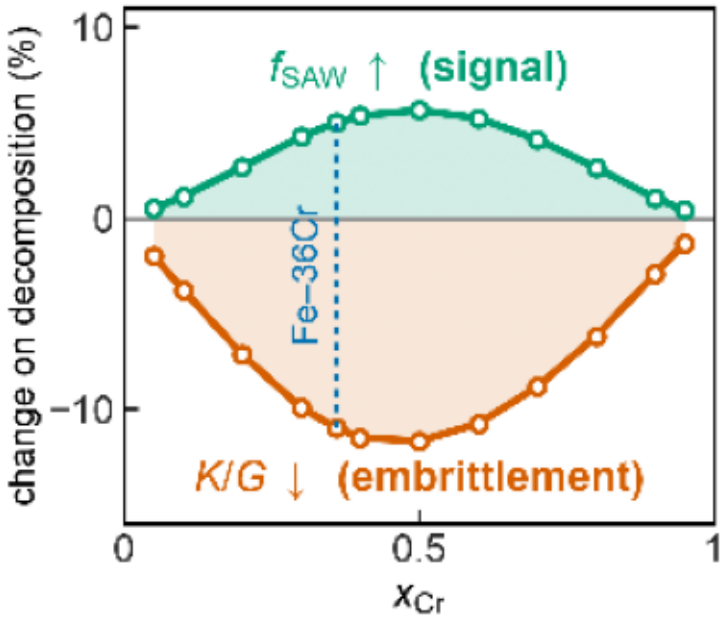



One spinodal band: the 748 K CALPHAD spinodal (30-71% Cr, Sec. 3.1) instead of the two 298 K PWSCF-PAW windows of Fig. 1b, whose narrow second window near 95% Cr was confusing; +5.7% under the peak

# v11



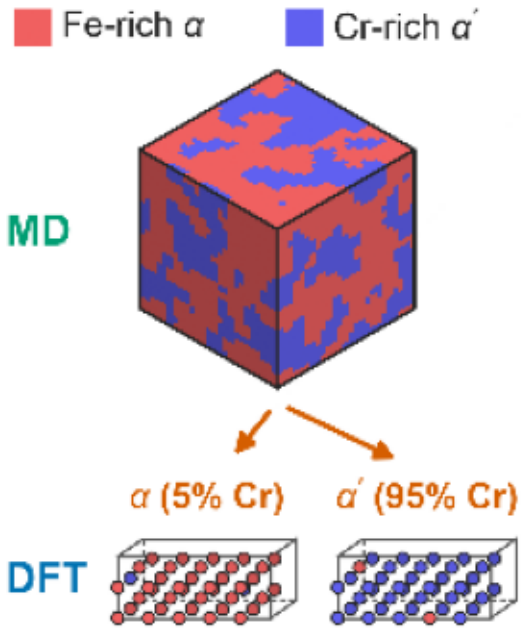

**Simulated spinodal decomposition of Fe–36Cr**

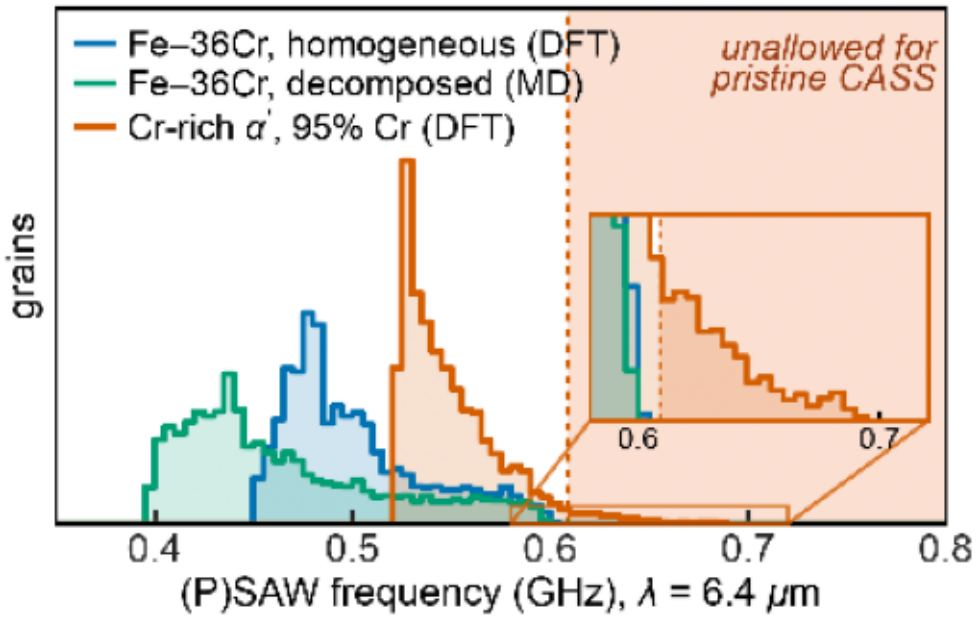
**DFT: stiff Cr-rich α′ places surface waves in the range unallowed for pristine ferrite**

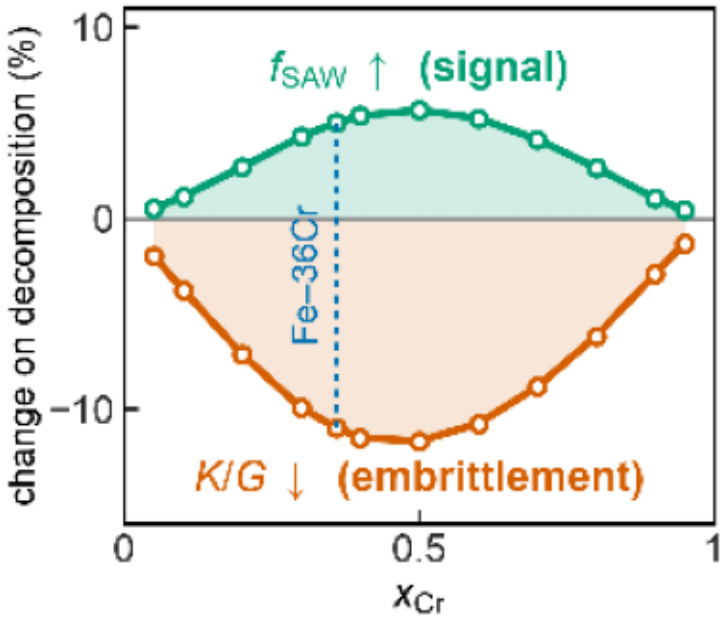
**Shear stiffening raises $f_{SAW}$ and lowers $K/G$: a signature of embrittlement**

New design after the Acta 2023 graphical abstract: three panels with bold claims beneath (MD decomposed Fe-36Cr cell; (P)SAW spectrum over grain orientations with inset on the unallowed range, DFT alpha' reaching 0.69 GHz; f_SAW up and K/G down at all twelve MD compositions); no spinodal shading

# v12



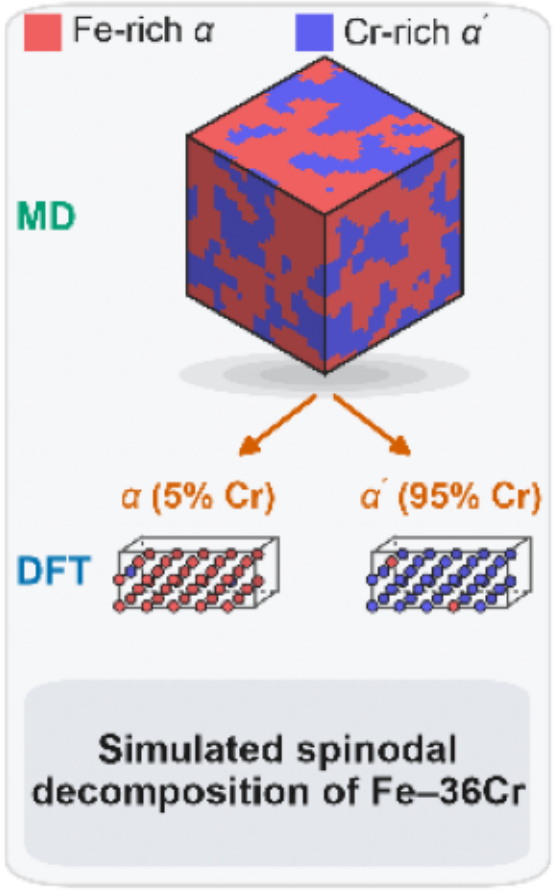

**Simulated spinodal decomposition of Fe–36Cr**

**DFT: stiff Cr-rich α′ places surface waves in the range unallowed for pristine ferrite**

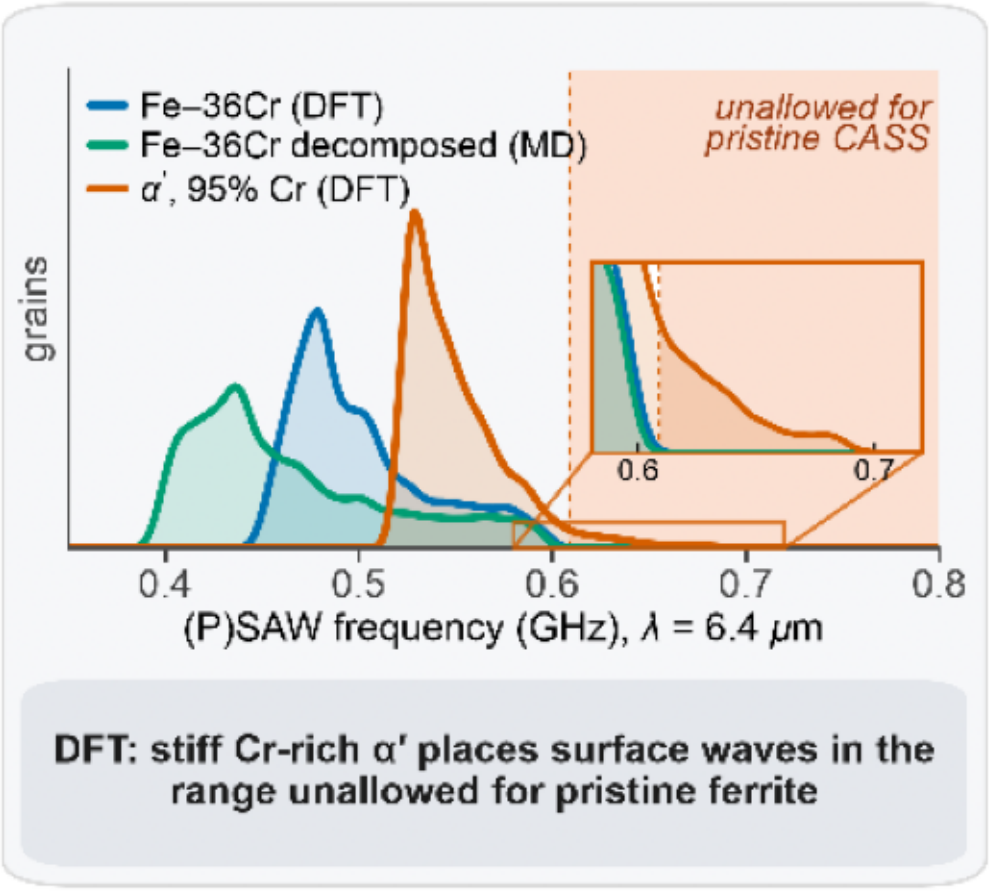
**Shear stiffening raises $f_{SAW}$ and lowers $K/G$: a signature of embrittlement**

v11 plus the QE 40-atom cells of alpha (5% Cr) and alpha' (95% Cr) under the MD cube in panel 1, with arrows from the cube (MD separates, DFT models the two phases)

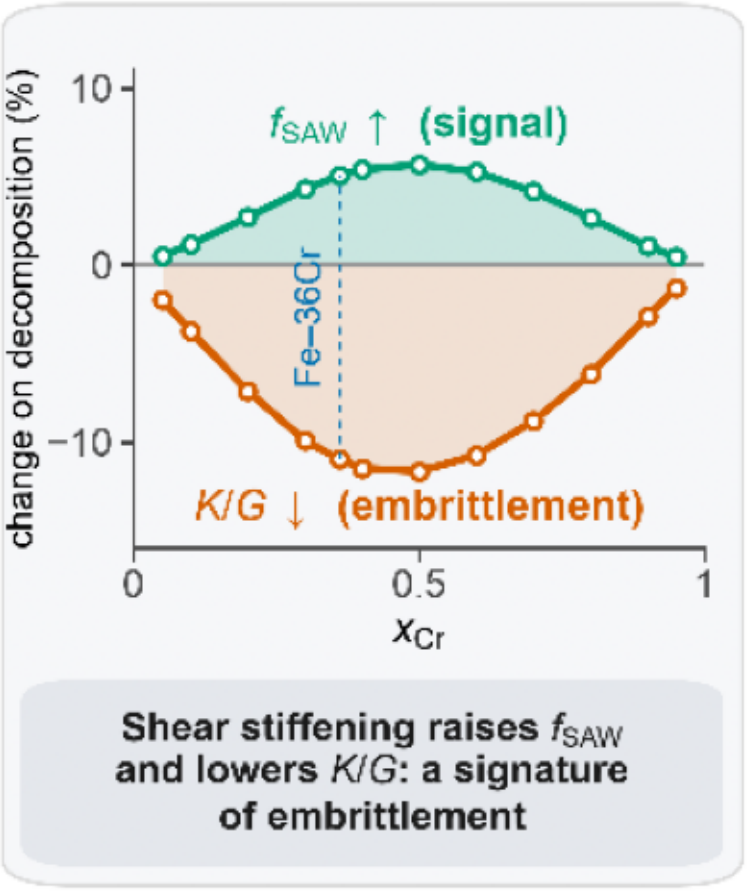
# v13




Simulated spinodal decomposition of Fe–36Cr



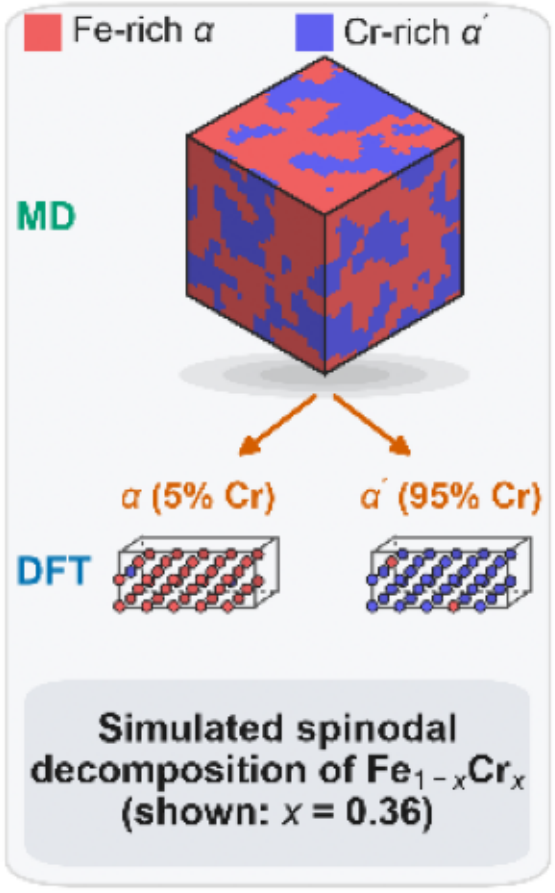

DFT: stiff Cr-rich α′ places surface waves in the range unallowed for pristine ferrite

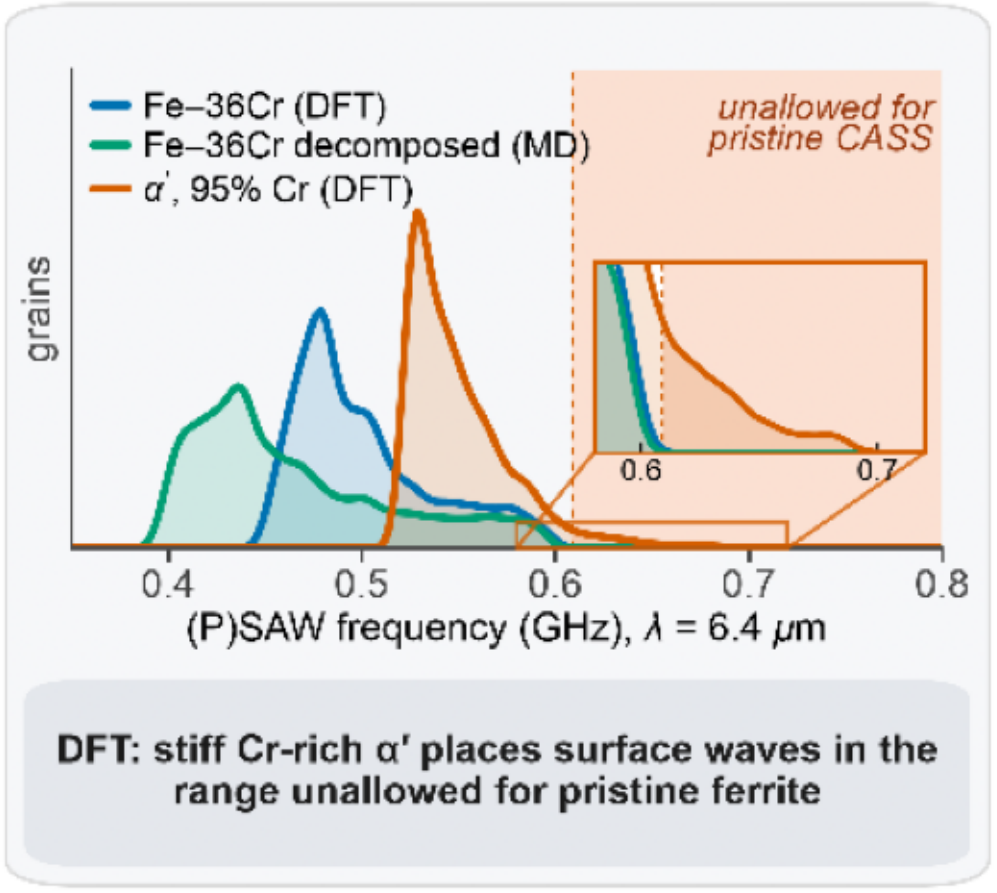


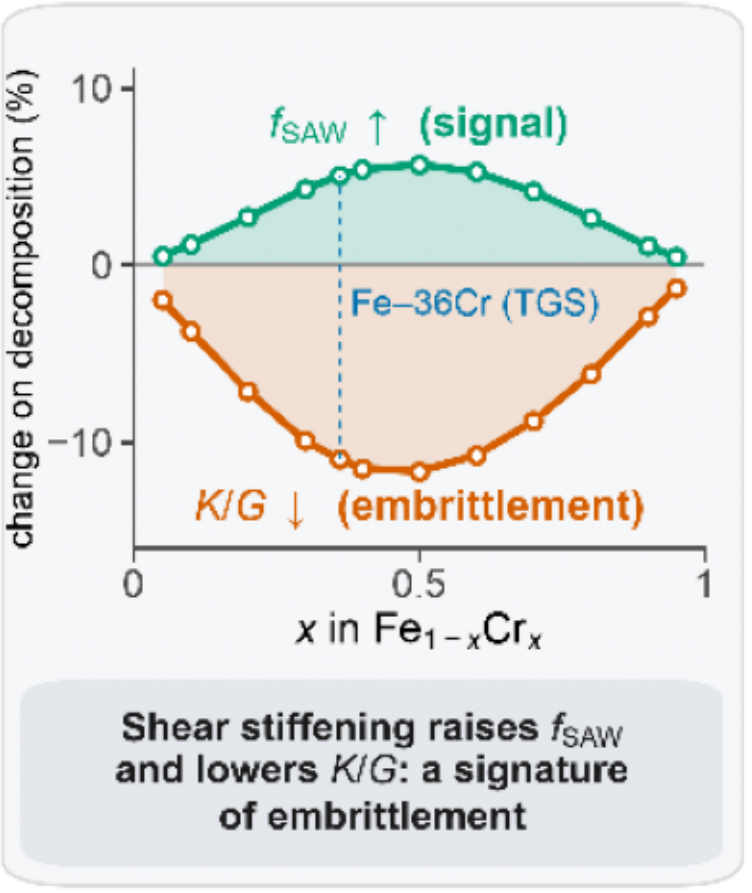

Shear stiffening raises $f_{SAW}$ and lowers $K/G$: a signature of embrittlement

v12 beautified, same content: rounded cards and claim strips, smoothed spectra, open axes, soft shadow under the cube, DFT cells symmetric about the cube, shorter legend

# v14

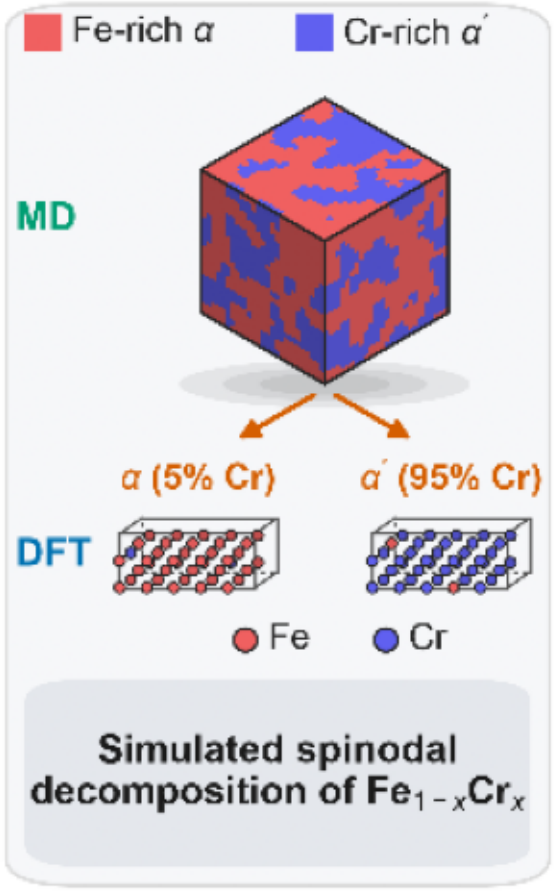


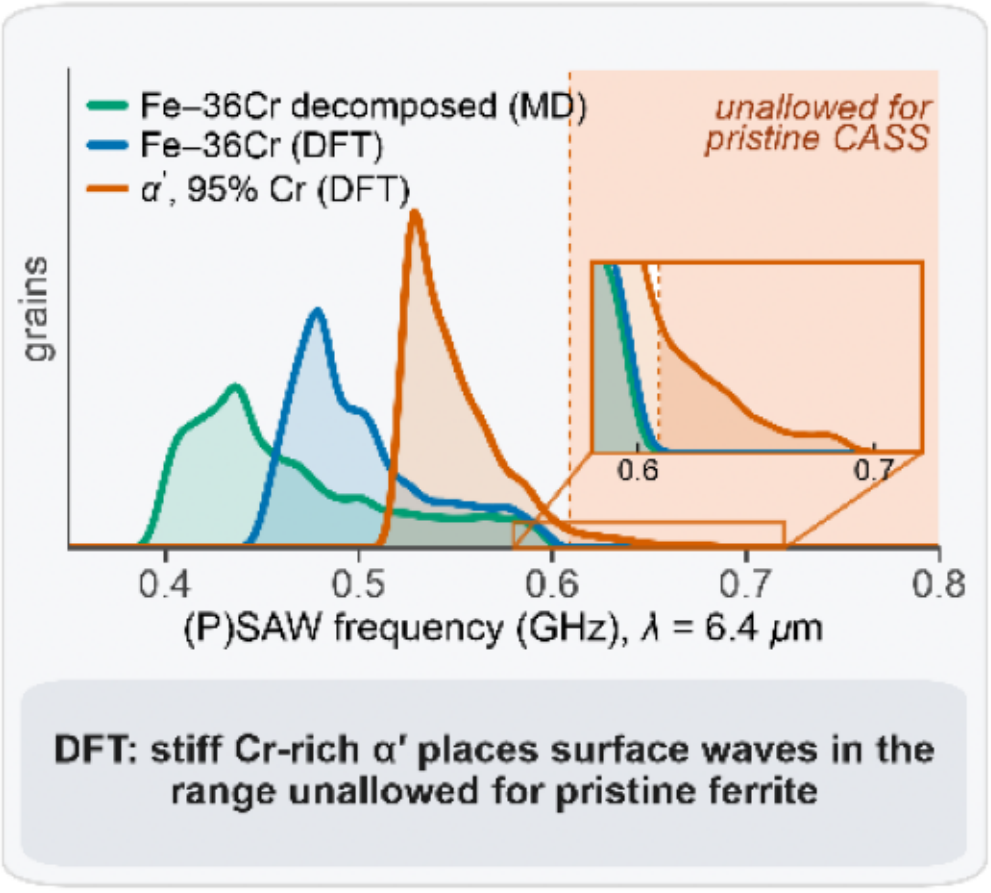



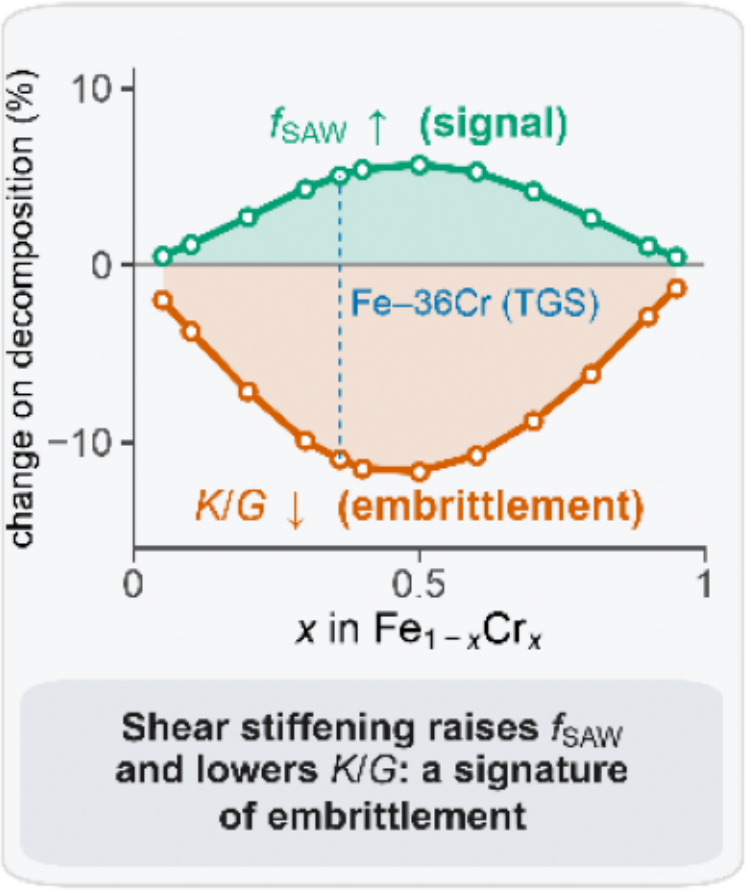




Fe(1-x)Cr(x) made explicit (panel 1 claim 'of Fe1-xCrx (shown: x = 0.36)', panel 3 axis 'x in Fe1-xCrx'); Fe-36Cr marked '(TGS)', the alloy in which TGS detected spinodal decomposition (Rae et al.), not a CASS composition

# v15




Centre legend ordered MD (green), DFT Fe-36Cr (blue), DFT alpha' (orange); panel 1 claim without Fe-36Cr ('Simulated spinodal decomposition of Fe1-xCrx'); cube smaller and clear of the key; Fe/Cr atom key under the DFT cells